

Advanced IT Ticket System with AI Agent – System Design

1. Core System Modules

Authentication System

Ticket Management

AI Support Agent

AMC Customer Management

Billing & Subscription

Work Tracking

Analytics Dashboard

2. User Roles

Admin:

Add AMC customers

Add monthly billing customers

View billing reports

Assign agents

Manage SLA

Analytics

Support Agent:

View assigned tickets

Update status

Add work logs

Upload files

Use AI assistant

Client / Customer:

Raise ticket

Upload screenshot

View ticket progress

Chat with AI assistant

3. AI Agent Function

AI assists before human support.

User describes the problem.

AI suggests solutions automatically.

If unresolved, the AI creates a ticket and assigns priority and category.

4. Smart AI Ticket Classification

AI automatically detects ticket category and priority.

Examples:

Server issue

Website bug

Login issue

Payment issue

Email issue

Priority detection:

Critical – server down

High – payment error

Medium – functional issue

Low – general support

5. Screenshot Analysis

When users upload screenshots, AI can detect:

Error codes

UI issues

Browser problems

Server errors

6. Ticket Creation Fields

Customer Name

Company

Issue Title

Description

Priority

Category

Project

Attachments

Screenshot upload

7. Work Tracking

Agents log work details:

Start time

End time

Description

Time spent

Attachments

8. AMC Customer Management

Admin can add Annual Maintenance Contract customers.

Fields include:

Company name

Contract start date

Contract end date

Monthly support hours

Priority SLA

Support hours are deducted when agents log time.

9. Monthly Billing Customers

Customers without AMC contracts are billed based on work hours.

Admin sets hourly rate and service cost.

Invoices are generated automatically.

10. Billing System

Admin-only visibility.

Agents and customers cannot see internal billing calculations.

Invoices can be exported as PDF or Excel reports.

11. SLA Management

Example SLA rules:

Critical – 30 minute response

High – 2 hours

Medium – 8 hours

Low – 24 hours

12. Admin Panel Modules

Dashboard

Tickets

Customers

AMC Contracts

Monthly Customers

Agents

Billing

Reports

Knowledge Base

System Settings

13. Database Tables

users

customers

tickets
ticket_comments
attachments
work_logs
contracts_amc
billing
invoices
knowledge_base
ai_logs

14. Technology Stack

Frontend: React.js, Next.js, Tailwind

Backend: Node.js, Express

AI: OpenAI API, LangChain

Database: PostgreSQL

Storage: AWS S3

Realtime: WebSockets / Socket.io

15. System Flow

Client submits problem

AI suggests solution

If unresolved, ticket created

Admin assigns agent

Agent logs work

System deducts AMC hours or adds billing

Admin generates invoice